

PROGRAMMING FOR DATA SCIENCE

Telco Customer Churn Analysis

Data Analysis using Pandas, NumPy and Matplotlib

Data heading	IBM Telco Customer Churn Dataset (7,043 customers)
Subject	Programming for Data Science
Faculty	Prof. _____
Student 1	Rishiraj Saha PRN: 26020845025
Student 2	Ritika Mondol PRN: 26020845026
Institute	Symbiosis Institute of Business Management
Date of submission	24 September 2026

Chapter 1: Introduction and Data Description

1.1 Introduction

Customer churn means customers stopping their service and leaving a company. For telecom companies this is a big problem, because getting a new customer usually costs a lot more than keeping an existing one. When a customer leaves, the company also loses all the monthly payments that customer would have made in the future.

In this project we looked at data from a telecom company to understand which customers are leaving, when they leave, and why. We used Python for the full analysis. Pandas was used to load, clean and summarise the data, NumPy for calculations like averages, percentiles and correlation, and Matplotlib to make charts. For each part we have shown the code, the output we got, and what we understood from it.

1.2 Objectives

- To find out what percentage of customers left the company and how much revenue was lost.
- To see which types of customers (by contract, internet service and payment method) churn the most.
- To check how churn changes with tenure, meaning how long the customer has been with the company.
- To look at the main reasons customers gave for leaving and suggest ways to reduce churn.

1.3 About the Dataset

We used the **Telco Customer Churn** dataset created by IBM. We downloaded it from Kaggle (kaggle.com/datasets/ylchang/telco-customer-churn-1113). It is about a made-up telecom company in California, USA, and has one row for each customer. The file we used is `Telco_customer_churn.xlsx`.

Detail	Value
Number of rows	7,043 customers
Number of columns	33
Categorical columns	Gender, Partner, Contract, Internet Service, Payment Method, Churn Label, etc.
Numerical columns	Tenure Months, Monthly Charges, Total Charges, CLTV, Churn Score
Target column	Churn Value (1 = customer left, 0 = customer stayed)
Source	IBM sample data, available on Kaggle

1.4 Important Columns

Column	Type	What it means
CustomerID	Text	Unique ID of each customer
City, Zip Code	Categorical	Where the customer lives
Gender, Senior Citizen, Partner, Dependents	Categorical	Basic details about the customer
Tenure Months	Numerical	How many months the customer has been with the company
Phone Service, Multiple Lines	Categorical	Phone plan details
Internet Service	Categorical	DSL, Fiber optic or no internet
Online Security, Tech Support, etc.	Categorical	Extra services taken by the customer
Contract	Categorical	Month-to-month, One year or Two year
Payment Method	Categorical	How the customer pays the bill
Monthly Charges	Numerical	Current monthly bill in dollars
Total Charges	Numerical	Total amount paid till now
Churn Value / Churn Label	Numerical / Categorical	Whether the customer left (1 / Yes) or not (0 / No)
Churn Score	Numerical	A score from 0 to 100 given by IBM showing chance of leaving
CLTV	Numerical	Customer Lifetime Value, i.e. expected value of the customer
Churn Reason	Categorical	The reason given by the customer for leaving

1.5 Tools Used

We used Python 3 with three libraries: Pandas, NumPy and Matplotlib. The code was written as a single Python file (telco_churn_analysis.py) and divided into small parts, one for each application.

Chapter 2: Data Analysis using Python

Getting Started: Importing Libraries and Loading the Data

Code

```
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4
5 pd.set_option("display.width", 120)
6 pd.set_option("display.max_columns", None)
7
8 df = pd.read_excel("../DataSet/Telco_customer_churn.xlsx")
9
10 print("Rows and columns:", df.shape)
11 print(df[["CustomerID", "Gender", "Tenure Months", "Contract",
12          "Monthly Charges", "Churn Label"]].head())
```

Output

```
Rows and columns: (7043, 33)
  CustomerID  Gender  Tenure Months  Contract  Monthly Charges  Churn Label
0  3668-QPYBK   Male           2  Month-to-month           53.85         Yes
1  9237-HQITU  Female           2  Month-to-month           70.70         Yes
2  9305-CDSKC  Female           8  Month-to-month           99.65         Yes
3  7892-P00KP  Female          28  Month-to-month          104.80         Yes
4  0280-XJGEX   Male          49  Month-to-month          103.70         Yes
```

Inference: The dataset loaded without any problem. It has 7,043 rows and 33 columns, so it easily meets the minimum of 510 rows. Each row is one customer, and the columns include categorical data (gender, contract, payment method) as well as numerical data (tenure, monthly charges, CLTV).

2.1 Pandas: Three Applications

Pandas 1 - Cleaning the data and basic statistics

Code

```
1 # Total Charges is stored as text, so we convert it to numbers
2 df["Total Charges"] = pd.to_numeric(df["Total Charges"], errors="coerce")
3
4 # checking missing values
5 print("Missing values in Total Charges:", df["Total Charges"].isnull().sum())
6 print("Missing values in Churn Reason:", df["Churn Reason"].isnull().sum())
7
8 # new customers have not paid anything yet, so we put 0
9 df["Total Charges"] = df["Total Charges"].fillna(0)
10
11 # removing columns we don't need
12 df = df.drop(columns=["Count", "Country", "State", "Lat Long"])
13 print("Shape after cleaning:", df.shape)
14
15 print(df[["Tenure Months", "Monthly Charges", "Total Charges", "CLTV"]].describe().round(2))
16
17 churn_rate = df["Churn Value"].mean() * 100
18 print("Overall churn rate:", round(churn_rate, 2), "%")
```

Output

```

Missing values in Total Charges: 11
Missing values in Churn Reason: 5174
Shape after cleaning: (7043, 29)

```

	Tenure	Months	Monthly Charges	Total Charges	CLTV
count	7043.00		7043.00	7043.00	7043.00
mean		32.37	64.76	2279.73	4400.30
std		24.56	30.09	2266.79	1183.06
min		0.00	18.25	0.00	2003.00
25%		9.00	35.50	398.55	3469.00
50%		29.00	70.35	1394.55	4527.00
75%		55.00	89.85	3786.60	5380.50
max		72.00	118.75	8684.80	6500.00

Overall churn rate: 26.54 %

Inference: Only two columns had missing values. Total Charges had 11 blanks, and when we checked, all of them were customers with 0 months of tenure who had not been billed yet, so we replaced them with 0. Churn Reason had 5,174 blanks, which is the same as the number of customers who stayed, so those blanks make sense and we left them. On average a customer has been with the company for about 32 months and pays around \$65 a month. The overall churn rate is 26.54%, so roughly 1 in every 4 customers left.

Pandas 2 - Churn rate by contract, internet service and payment method

Code

```

1 contract_churn = df.groupby("Contract")["Churn Value"].mean() * 100
2 print("Churn % by Contract")
3 print(contract_churn.round(2))
4
5 internet_churn = df.groupby("Internet Service")["Churn Value"].mean() * 100
6 print("\nChurn % by Internet Service")
7 print(internet_churn.round(2))
8
9 payment_churn = df.groupby("Payment Method")["Churn Value"].mean() * 100
10 print("\nChurn % by Payment Method")
11 print(payment_churn.round(2))

```

Output

```

Churn % by Contract
Contract
Month-to-month    42.71
One year          11.27
Two year           2.83
Name: Churn Value, dtype: float64

Churn % by Internet Service
Internet Service
DSL              18.96
Fiber optic     41.89
No               7.40
Name: Churn Value, dtype: float64

Churn % by Payment Method
Payment Method
Bank transfer (automatic)    16.71
Credit card (automatic)    15.24
Electronic check            45.29
Mailed check                19.11
Name: Churn Value, dtype: float64

```

Inference: Contract type makes the biggest difference. Customers on a month-to-month plan churned at 42.7%, while only 2.8% of two-year customers left. Fiber optic users churned at almost 42%, which is more than double the DSL rate. This was surprising because fiber is the premium service. Customers paying by electronic

check also churned a lot more (45.3%) than those using automatic bank transfer or credit card (15 to 17%).

Pandas 3 - Tenure groups, pivot table and churn reasons

Code

```
1 df["Tenure Group"] = pd.cut(df["Tenure Months"], bins=[-1, 12, 24, 48, 72],
2                             labels=["0-12", "13-24", "25-48", "49-72"])
3
4 table = pd.pivot_table(df, values="Churn Value", index="Tenure Group",
5                         columns="Contract", aggfunc="mean", observed=False)
6 print("Churn % by Tenure Group and Contract")
7 print((table * 100).round(1))
8
9 churned = df[df["Churn Value"] == 1]
10 print("\nTop 8 reasons for churn")
11 print(churned["Churn Reason"].value_counts().head(8))
```

Output

```
Churn % by Tenure Group and Contract
Contract      Month-to-month  One year  Two year
Tenure Group
0-12              51.4         10.5      0.0
13-24             37.7          8.1      0.0
25-48             32.9         10.6      2.2
49-72             26.0         12.9      3.3

Top 8 reasons for churn
Churn Reason
Attitude of support person          192
Competitor offered higher download speeds  189
Competitor offered more data         162
Don't know                          154
Competitor made better offer          140
Attitude of service provider         135
Competitor had better devices         130
Network reliability                   103
Name: count, dtype: int64
```

Inference: The pivot table shows that new customers on month-to-month plans are the most likely to leave: more than half of them (51.4%) churned in their first year. None of the two-year contract customers left in their first 24 months. When we looked at the reasons, the single biggest one was the attitude of the support person (192 customers). However, if we add up all the competitor-related reasons (better speeds, more data, better offer, better devices), they come to more than 600 customers, so competition is the main problem overall.

2.2 NumPy: Three Applications

NumPy 1 - Comparing monthly charges of churned and retained customers

Code

```
1 churn_yes = df[df["Churn Value"] == 1]["Monthly Charges"].values
2 churn_no = df[df["Churn Value"] == 0]["Monthly Charges"].values
3
4 print("Churned customers:", len(churn_yes))
5 print("Mean:", np.round(np.mean(churn_yes), 2))
6 print("Median:", np.round(np.median(churn_yes), 2))
7 print("Std deviation:", np.round(np.std(churn_yes), 2))
8 print("25th and 75th percentile:", np.percentile(churn_yes, [25, 75]))
9
10 print("\nRetained customers:", len(churn_no))
11 print("Mean:", np.round(np.mean(churn_no), 2))
12 print("Median:", np.round(np.median(churn_no), 2))
13 print("Std deviation:", np.round(np.std(churn_no), 2))
14 print("25th and 75th percentile:", np.percentile(churn_no, [25, 75]))
15
16 print("\nDifference in mean:", np.round(np.mean(churn_yes) - np.mean(churn_no), 2))
```

Output

```
Churned customers: 1869
Mean: 74.44
Median: 79.65
Std deviation: 24.66
25th and 75th percentile: [56.15 94.2 ]

Retained customers: 5174
Mean: 61.27
Median: 64.43
Std deviation: 31.09
25th and 75th percentile: [25.1 88.4]

Difference in mean: 13.18
```

Inference: Customers who left were paying more. Their average monthly bill was \$74.44, compared to \$61.27 for customers who stayed, a gap of about \$13. The median shows the same pattern (\$79.65 vs \$64.43). The retained group has a lower 25th percentile (\$25.10) because many of them are on cheap phone-only plans. So higher monthly charges seem to go along with a higher chance of leaving.

NumPy 2 - Correlation between numeric columns

Code

```
1 cols = ["Tenure Months", "Monthly Charges", "Total Charges", "CLTV", "Churn Value"]
2 data = df[cols].values
3
4 corr = np.corrcoef(data.T)
5 print(pd.DataFrame(np.round(corr, 2), index=cols, columns=cols))
```

Output

	Tenure Months	Monthly Charges	Total Charges	CLTV	Churn Value
Tenure Months	1.00	0.25	0.83	0.40	-0.35
Monthly Charges	0.25	1.00	0.65	0.10	0.19
Total Charges	0.83	0.65	1.00	0.34	-0.20
CLTV	0.40	0.10	0.34	1.00	-0.13
Churn Value	-0.35	0.19	-0.20	-0.13	1.00

Inference: Tenure has the strongest link with churn among these columns (-0.35). The negative sign means that the longer someone stays, the less likely they are to leave. Monthly charges have a small positive link (0.19), which matches what we

found in NumPy 1. Tenure and Total Charges are strongly related (0.83), but that is expected since total charges build up over time.

NumPy 3 - Risk groups and revenue lost

Code

```
1 score = df["Churn Score"].values
2 churn = df["Churn Value"].values
3 monthly = df["Monthly Charges"].values
4
5 risk = np.where(score >= 80, "High", np.where(score >= 50, "Medium", "Low"))
6
7 for group in ["High", "Medium", "Low"]:
8     customers = np.sum(risk == group)
9     rate = np.mean(churn[risk == group]) * 100
10    print(group, "risk:", customers, "customers, churn rate =", round(rate, 1), "%")
11
12 lost = np.sum(monthly[churn == 1])
13 total = np.sum(monthly)
14 print("\nMonthly revenue lost: $", round(lost, 2))
15 print("Share of total revenue lost:", round(lost / total * 100, 1), "%")
16 print("Yearly revenue lost: $", round(lost * 12, 2))
```

Output

```
High risk: 1201 customers, churn rate = 92.0 %
Medium risk: 3309 customers, churn rate = 23.1 %
Low risk: 2533 customers, churn rate = 0.0 %

Monthly revenue lost: $ 139130.85
Share of total revenue lost: 30.5 %
Yearly revenue lost: $ 1669570.2
```

Inference: We used np.where to put customers into High, Medium and Low risk groups based on their churn score. The High risk group really did leave: 92% of them churned, while nobody in the Low group did. So the score is useful for deciding who to contact first. The company is losing about \$139,131 in monthly revenue because of churn. That is 30.5% of all monthly revenue, or about \$1.67 million a year. The revenue share (30.5%) is higher than the customer share (26.5%) because the customers who leave usually pay more.

2.3 Matplotlib: Three Applications

Matplotlib 1 - Overall churn and churn by contract

Code

```
1 plt.figure(figsize=(11, 4))
2
3 plt.subplot(1, 2, 1)
4 counts = df["Churn Label"].value_counts()
5 plt.pie(counts, labels=["Stayed", "Churned"], autopct="%1.1f%%",
6         colors=["steelblue", "salmon"], startangle=90)
7 plt.title("Overall Churn")
8
9 plt.subplot(1, 2, 2)
10 plt.bar(contract_churn.index, contract_churn.values, color=["salmon", "gold", "mediumseagreen"])
11 plt.title("Churn % by Contract Type")
12 plt.ylabel("Churn %")
13
14 plt.tight_layout()
15 plt.show()
```

Chart

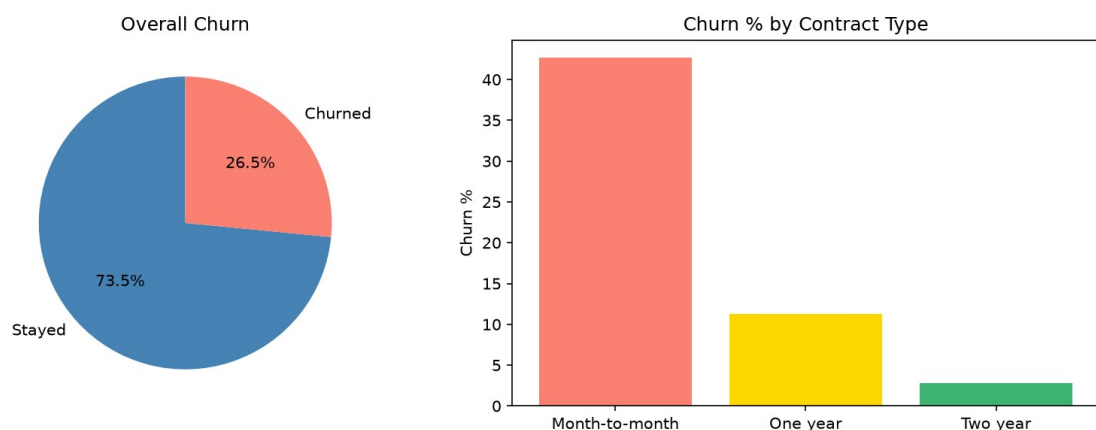


Figure 1: Overall churn and churn by contract

Inference: The pie chart shows the overall split of 73.5% stayed and 26.5% churned. The bar chart makes the effect of contract type very clear: the month-to-month bar is far taller than the other two. Moving customers to one-year or two-year contracts looks like the easiest way to cut churn.

Matplotlib 2 - Monthly charges and tenure of churned vs retained customers

Code

```

1 plt.figure(figsize=(11, 4))
2
3 plt.subplot(1, 2, 1)
4 plt.hist(churn_no, bins=30, alpha=0.6, label="Stayed", color="steelblue")
5 plt.hist(churn_yes, bins=30, alpha=0.6, label="Churned", color="salmon")
6 plt.title("Monthly Charges")
7 plt.xlabel("Monthly Charges ($)")
8 plt.ylabel("Number of customers")
9 plt.legend()
10
11 plt.subplot(1, 2, 2)
12 tenure_no = df[df["Churn Value"] == 0]["Tenure Months"]
13 tenure_yes = df[df["Churn Value"] == 1]["Tenure Months"]
14 plt.boxplot([tenure_no, tenure_yes])
15 plt.xticks([1, 2], ["Stayed", "Churned"])
16 plt.title("Tenure of Customers")
17 plt.ylabel("Tenure (months)")
18
19 plt.tight_layout()
20 plt.show()

```

Chart

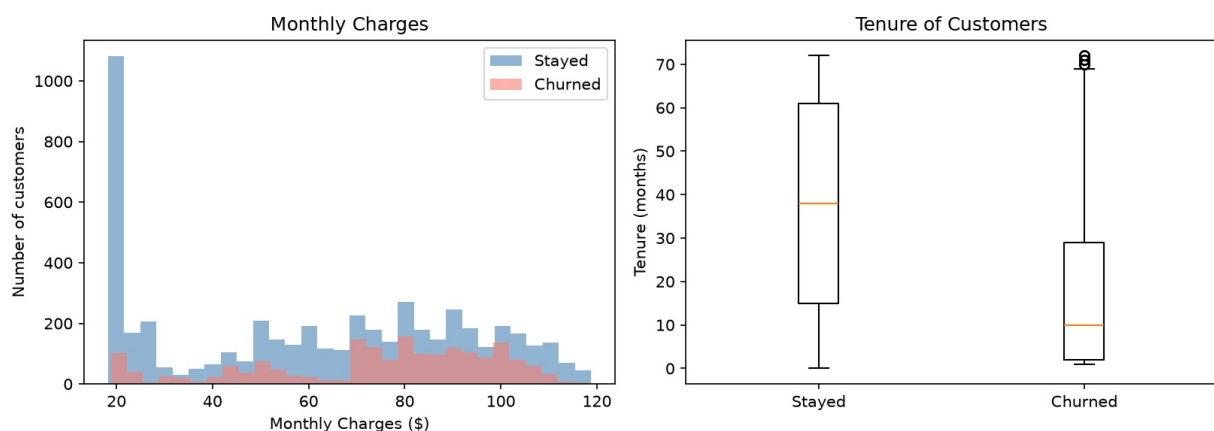


Figure 2: Monthly charges and tenure of churned vs retained customers

Inference: In the histogram, a lot of the customers who stayed are around \$20 a month, while churned customers are bunched between \$70 and \$105. That is the price range of fiber and streaming plans. The box plot shows that churned customers usually leave early. Their median tenure is only about 10 months, compared to about 38 months for customers who stayed.

Matplotlib 3 - Churn over tenure and top churn reasons

Code

```

1 plt.figure(figsize=(12, 4.5))
2
3 plt.subplot(1, 2, 1)
4 tenure_churn = df.groupby("Tenure Months")["Churn Value"].mean() * 100
5 plt.plot(tenure_churn.index, tenure_churn.values, color="salmon")
6 plt.title("Churn % by Tenure")
7 plt.xlabel("Tenure (months)")
8 plt.ylabel("Churn %")
9 plt.grid(alpha=0.3)
10
11 plt.subplot(1, 2, 2)
12 reasons = churned["Churn Reason"].value_counts().head(8)
13 plt.barh(reasons.index, reasons.values, color="orange")
14 plt.gca().invert_yaxis()
15 plt.title("Top 8 Churn Reasons")
16 plt.xlabel("Number of customers")
17
18 plt.tight_layout()
19 plt.show()

```

Chart

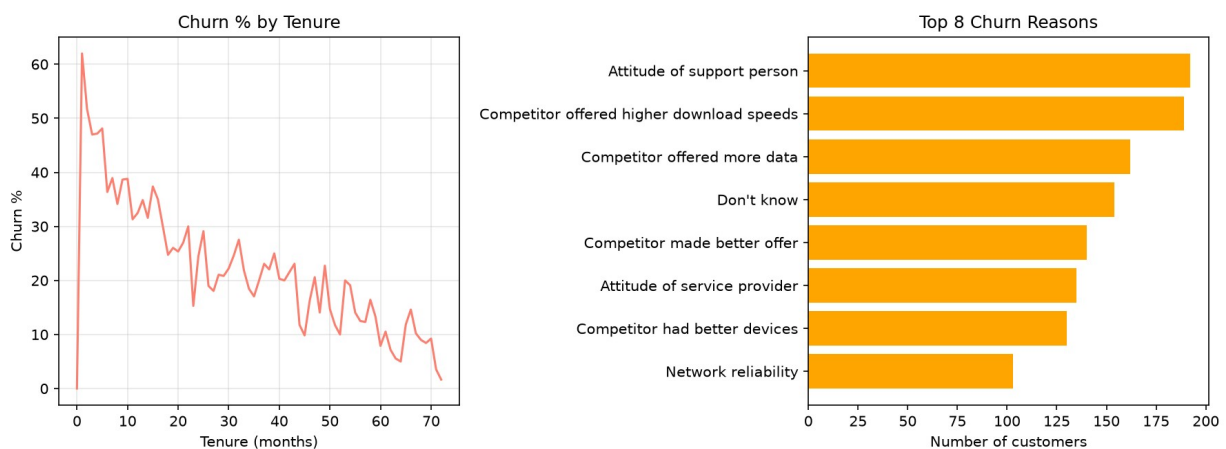


Figure 3: Churn over tenure and top churn reasons

Inference: The line chart shows churn is very high in the first few months (around 50 to 60%) and then keeps going down, reaching below 10% for customers who have been around for 5 to 6 years. The value at 0 months is 0% only because those customers joined very recently. The bar chart of reasons shows two main themes: problems with service or staff attitude, and better offers from competitors.

Chapter 3: Conclusion

3.1 What the Data Tells Us

About one out of every four customers (26.5%) left the company. The customers who left were paying more than average, so the company lost 30.5% of its monthly revenue, which comes to around \$1.67 million a year. The churn is not spread evenly across customers. Most of it comes from a few clear groups, as shown in the table below.

Factor	Churns the most	Churns the least
Contract	Month-to-month (42.7%)	Two year (2.8%)
Tenure	First 12 months (47.4%)	49 to 72 months (9.5%)
Internet service	Fiber optic (41.9%)	No internet (7.4%)
Payment method	Electronic check (45.3%)	Credit card, automatic (15.2%)
Monthly charges	Churned: \$74.44 on average	Stayed: \$61.27 on average

3.2 Key Findings

- **Contract type matters the most.** Customers on longer contracts hardly ever leave, even when they are new.
- **The first year is the riskiest time.** Churn is highest in the first few months and keeps falling as customers stay longer. Most customers who leave do so within about 10 months.
- **Fiber optic customers are not happy.** They pay the most but churn twice as much as DSL users. Many customers also said competitors offered better speed or more data.
- **Payment method is linked to churn.** People paying by electronic check leave about three times more often than people on automatic payments.
- **Customer service is a real issue.** The single most common reason for leaving was the attitude of the support person.
- **Churn can be predicted.** The churn score correctly picks out a high risk group where 92% of customers left.

3.3 Suggestions

- Give discounts or small rewards to customers who move from month-to-month to a one-year or two-year contract.
- Pay extra attention to new customers in their first 3 to 6 months, for example with welcome calls and help with setup.
- Compare fiber prices and speeds with competitors, and think about adding free extras like online security or tech support.
- Encourage customers to switch to automatic payments by offering a small discount on the bill.
- Train support staff better and track customer satisfaction regularly.
- Use the churn score to contact high risk customers before they decide to leave.

3.4 Limitations

This dataset is from a made-up company and only covers one quarter in one US state, so the results may not exactly match a real company. Also, our analysis shows which things are linked to churn, but it does not prove that they cause it. In the future, a machine learning model could be built on this data to predict churn more accurately.